

Methods

1. TEST

8 held-out semiprimes

Uniform and
discrete-Gaussian Regev
samples

$M = 8, 16, 32$

64 repeats per setting

2. ADD CONTROLLED ERROR

Regev primary test:

omit 1–2 smallest QFT
phase layers

Regev stress tests:

1% readout flips, 5%
corrupted shots, and
bounded noisy-dual
displacement

Shor companion tests:

phase/RZ rotation error
and 0.1–1% CX depolariz-
ing noise

3. RECOVER

Samples

↓

augmented integer lattice

↓

LLL reduction

↓

verify relation + GCD factor

4. MEASURE

Factor-recovery success

Quantum samples required

Controlled-phase gates

CX gates and circuit depth

LLL runtime

Important: these are controlled simulations. QFT truncation is

